

Langlin Huang

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EDUCATION

- **Washington University in St. Louis (WashU)** St. Louis, U.S.
Ph.D. Candidate in Computer Science - GPA: 4.0/4.0 Aug. 2024 - current
- **Institute of Computing Technology, Chinese Academy of Sciences (ICT/CAS)** Beijing, China
M.E. in Computer Technology - GPA: 3.75/4.0, Outstanding Dissertation Sep. 2021 - Jun. 2024
- **University of International Business and Economics (UIBE)** Beijing, China
B.E. in Data Science and Big Data Technology - GPA: 3.74/4.0 (Rank:2/147) Sep. 2017 - Jun. 2021

RESEARCH INTERESTS

- Agentic RL and On-Policy Distillation
- Large Language Model Reasoning & Post-Training
- Large Language Model Efficiency

PUBLICATIONS

- **On the Off-Policy Teacher in On-Policy Distillation**
Langlin Huang*, Hao Liu*, Mononito Goswami, Xinyu Li, Prithwish Jana, Nikos Kanakaris, Patrick Blöbaum, Purak Jain
NeurIPS 2026 workshops: MATH-AI and Pre-to-Post [Paper] [Project]
- **Nonsense Helps: Prompt Space Perturbation Broadens Reasoning Exploration**
Langlin Huang, Chengsong Huang, Jinyuan Li, Donghong Cai, Yuyi Yang, Jiaxin Huang
EMNLP 2026 [Paper] [Code]
- **SPEED: Specialized Position Experts for Efficient Speculative Decoding**
Langlin Huang, Chengsong Huang, Jixuan Leng, Di Huang, Jiaxin Huang
COLM 2026 [Paper] [Code]
- **G-Zero: Self-Play for Open-Ended Generation from Zero Data**
Chengsong Huang, Haolin Liu, Tong Zheng, Runpeng Dai, **Langlin Huang**, Jinyuan Li, Zongxia Li, Zhepei Wei, Yu Meng, Jiaxin Huang
NeurIPS 2026 spotlight [Paper] [Code]
- **Training Data Efficiency in Multimodal Process Reward Models**
Jinyuan Li, Chengsong Huang, **Langlin Huang**, Shaoyang Xu, Haolin Liu, Wenxuan Zhang, Jiaxin Huang
ICML 2026 [Paper][Code]
- **CaTS: Calibrated Test-Time Scaling for Efficient LLM Reasoning**
Chengsong Huang, **Langlin Huang**, Jixuan Leng, Jiacheng Liu, Jiaxin Huang
ICLR 2026 [Paper][Code]
- **Divide, Reweight, and Conquer: A Logit Arithmetic Approach for In-Context Learning**
Chengsong Huang, **Langlin Huang**, Jiaxin Huang
EACL 2026 oral [Paper][Code]
- **RelayLLM: Efficient Reasoning via Collaborative Decoding**
Chengsong Huang, Tong Zheng, **Langlin Huang**, Jinyuan Li, Haolin Liu, Jiaxin Huang
EMNLP 2026 [Paper]
- **CrossWordBench: Evaluating the Reasoning Capabilities of LLMs and LVLMs with Controllable Puzzle Generation**
Jixuan Leng, Chengsong Huang, **Langlin Huang**, Bill Yuchen Lin, William W Cohen, Haohan Wang, Jiaxin Huang
COLM 2025 [Paper][Code] [Dataset]
- **MoCE: Adaptive Mixture of Contextualization Experts for Byte-based Neural Machine Translation**
Langlin Huang, Mengyu Bu, Yang Feng
NAACL 2025 oral [Paper] [Code]
- **BayLing: Bridging Cross-lingual Alignment and Instruction Following through Interactive Translation for Large Language Models**
Shaolei Zhang, Qingkai Fang, Zhuocheng Zhang, Zhengrui Ma, Yan Zhou, **Langlin Huang**, Mengyu Bu, Shangdong Gui, Yunji Chen, Xilin Chen, Yang Feng
Technical Report [Paper] [Code]
- **Integrating Multi-scale Contextualized Information for Byte-based Neural Machine Translation**
Langlin Huang, Yang Feng
ACL findings, 2024 [Paper][Code]
- **Enhancing Neural Machine Translation with Semantic Units**
Langlin Huang, Shuhao Gu, Zhuocheng Zhang, Yang Feng
EMNLP findings, 2023 [Paper][Code]
- **Automatic Construction of a Depression-Domain Lexicon Based on Microblogs: Text Mining Study**
Genghao Li, Bing Li, **Langlin Huang**, Sibing Hou
JMIR medical informatics, 2020, Vol 8. Jun. 2020 [Paper]

COMMUNITY SERVICES

Top-reviewer: NeurIPS 2025

Reviewer: NeurIPS, ICML, ICLR, COLM, ACL, ACL Rolling Review, TALLIP

INDUSTRY EXPERIENCE

•AWS Agentic AI DevOps, Amazon

May. 2026 - Sep. 2026

Applied Scientist Intern

Identified the **off-policy teacher issue** in On-Policy Distillation (OPD). Proposed updating the teacher policy conditioned on the student prefixes, improving the student's performance on math reasoning and code generation.

- We found that even a stronger model performs badly when continuing from a worse model-generated prefix, proving its inadequacy to be a good teacher.
- We proposed a strategy named Student-CONditioned Update of the Teacher (SCOUT), which effectively alleviates the mismatch between teacher and student models.
- We empirically prove that sparse teacher-side updates are sufficient to mitigate the off-policy teacher issue and improve the final performance.

•Search Relevance Team, ByteDance

Jan. 2024 - Apr. 2024

Algorithm Engineer

Proposed an adapter module to bridge the relevance model used in production with an LLM. The adapter transformed representations from the traditional model into LLM-compatible hidden states, enabling:

- Joint utilization of pretrained traditional models and powerful LLM reasoning capabilities.
- Robust handling of out-of-vocabulary and multilingual inputs.
- A representation model plus LLM framework extensible to user-isolated data or multimodal input scenarios.